Composition, Automorphism and Fractal Fixed-Point Tests

The Identity-or-Erasure Theorem and the Minimal Ternary Instrument

FIN ST2262–ST2351

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Six adaptive research rounds — 90 programmes
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Release 11.08 found the strict loop-product flag lift but left two possible routes to uniqueness: separate edge linearity plus face universality, or a fractal refinement principle selecting the Hodge exponent. This report tests both routes, proves an identity-or-erasure theorem for exact finite self-similarity, and constructs the smallest instrument capable of observing irreducible three-point information.

Abstract

The strict loop-product field
$$\tau_{ijk} = W_{ij}W_{jk}W_{ki}$$
supports the positive Hodge family
$$h_f(p) = \frac{\tau_f^p}{\langle \tau^p \rangle}, \qquad p > 0.$$
Release 11.09 asks whether the exponent $p = 1$ can be forced without inserting new physics.

The first route fails. Dirichlet energy is exactly additive under parallel addition of edge conductances, but this does not determine how three edge weights compose into one face weight. The product is separately additive, whereas the geometric mean, harmonic mean and minimum are equally local, positive, symmetric and support-zero but are not separately additive. Separate edge linearity is therefore an additional face-lift axiom.

The second premise, face universality, also fails. All six strict cyclic-distance weights are distinct. The maximum-weight edges form C_{12}, so every weighted automorphism belongs to $\mathrm{Aut}(C_{12}) = D_{12}$; conversely D_{12} preserves every weight. Hence
$$\mathrm{Aut}(W) = D_{12}.$$
This group has twelve triangle orbits, all distinguished by the strict loop product. Equivariance permits twelve face coefficients rather than one.

A stronger conditional result does exist. For a positive profile define
$$T_p(x) = \frac{x^p}{\langle x^p \rangle}.$$
Then
$$T_p \circ T_q = T_{pq}.$$
For any nonconstant positive profile, scale-stationary path independence $T_p^2 x = T_p x$ implies $p^2 = p$. Thus only $p = 0$ and $p = 1$ survive. The first erases the profile to uniformity; the second is the identity. Even self-similarity up to a permutation of the twelve orbits forces $p = 1$. Consequently exact finite power self-similarity supplies no nontrivial compression dynamics.

Ordinary coassociativity is weaker and leaves a continuous rate: $p(t) = e^{-ct}$, or one exponent per refinement level. This agrees with the existing FIN refinement theorems, which leave unsourced vertical rates. The conditional $p = 1$ theorem therefore cannot be promoted to strict FIN.

Finally, the minimal observable for the hidden-synergy family is $Y = x_1x_2x_3$. Pair-only Fisher information is zero, whereas parity has
$$I(\varepsilon) = \frac{1}{1 - \varepsilon^2}.$$
The estimator $\widehat{\varepsilon} = \overline{Y}$ is efficient. Nineteen events suffice under a Hoeffding bound to determine the sign for $|\varepsilon| \geq 0.75$ with declared error 0.01. A minimal non-Gaussian model is
$$p_\theta(x) = \frac{e^{\theta x_1 x_2 x_3}}{8\cosh\theta}, \qquad \varepsilon = \tanh\theta.$$
It explicitly adds the absent three-body statistic and source parameter.

The gate contains seven valid mathematical constructions, three strict-source rows, and zero physical rows. The pair/refinement route is a bounded no-go on current artifacts. A future result can change the verdict only by exporting a strict non-Gaussian three-body state law with nonzero connected cumulant, fixed polarity and coherent refinement transport.

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1 Input discipline

The report retains the strict/legacy split. It uses the strict positive Dirichlet conductances W in

$$A = sI - W, \qquad \langle f, Af\rangle = \frac{1}{2}\sum_{x,y} W_{xy}|f_x - f_y|^2.$$

No legacy physical role is transferred. The raw legacy loop products have both signs, so noninteger powers are not real without an additional absolute value, sign sector or completion rule.

The question is not whether $p = 1$ is an elegant convention. It is whether the current mathematics makes every alternative impossible.

2 Round I: ST2262–ST2276 — separate linearity

For fixed f, the Dirichlet energy is linear in the edge conductance matrix:

$$E_{W+W'}(f) = E_W(f) + E_{W'}(f).$$

The finite replay gives residual zero. This is an edge-level statement.

Theorem 1 (conditional separate-additivity theorem)**.** *Let $F(a, b, c)$ be continuous, vanish when any argument vanishes, and be additive independently in each positive argument. Then*

$$F(a, b, c) = \kappa abc.$$

Thus independent preservation of parallel addition selects the loop product. The key point is that FIN has not exported this preservation law for the edge-to-face lift.

The following target-free counterrules satisfy locality, positivity, symmetry and support-zero:

$$(abc)^{1/3}, \qquad \frac{3}{a^{-1} + b^{-1} + c^{-1}}, \qquad \min(a, b, c).$$

On a fixed strict weight tuple their separate-additivity defects are, respectively,

$$-0.0785221321, \quad -0.0785966273, \quad -0.0470291687,$$

while the product residual is numerical zero.

There is also a homogeneity ambiguity. The product has total degree three; the three counterrules have degree one. Current strict data do not assign a physical dimension or scaling degree to a face stiffness. Edge-energy linearity therefore does not force face multilinearity.

3 Round II: ST2277–ST2291 — face universality

The six strict distance weights are distinct, with minimum pair separation 0.0130604060. Distance-one edges have the unique maximum and form the cycle C_{12}.

Theorem 2 (weighted automorphism group)**.**

$$\mathrm{Aut}(W) = D_{12}.$$

Proof. Every weighted automorphism preserves the unique maximum-weight subgraph, which is C_{12}, so it belongs to $\mathrm{Aut}(C_{12}) = D_{12}$. Conversely every dihedral symmetry preserves cyclic distance and all six weights. □

The direct finite conjugation residual is zero for all 24 elements. A non-dihedral transposition gives residual 0.9178297, and multiplication by the unit $5 \in \mathbb{Z}_{12}$ also fails because it permutes unequal distance weights.

There are twelve D_{12}-orbits of triangles. The twelve loop-product values are distinct; their smallest separation is 2.5670×10^{-7}. The strict data therefore retain, rather than erase, the orbit labels.

Full S_{12} symmetry would make every triangle equivalent, but it is not a symmetry of the strict weighted graph. One universal formula on all weighted graphs is a valid category-level axiom, not an internal consequence of $\mathrm{Aut}(W)$.

4 Round III: ST2292–ST2306 — conditional exponent theorem

Theorem 3 (continuous multiplicative valuations)**.** *Every continuous positive solution of*

$$g(xy) = g(x)g(y)$$

has $g(x) = x^p$*. Monotonicity gives* $p \geq 0$*.*

After mean normalization,

$$T_p(T_q(x)) = T_{pq}(x).$$

The strict orbit-vector replay gives residual 3.55×10^{-15}.

Theorem 4 (idempotent exponent classification)**.** *Let* x *be positive and nonconstant. If*

$$T_p(T_p(x)) = T_p(x),$$

then $p \in \{0, 1\}$*.*

Proof. Choose $x_i \neq x_j$. Ratios remove normalization:

$$\left(\frac{x_i}{x_j}\right)^{p^2} = \left(\frac{x_i}{x_j}\right)^{p}.$$

Since the base ratio is positive and not one, $p^2 = p$. □

The $p = 0$ branch maps everything to the uniform profile and erases all orbit information. The $p = 1$ branch preserves information but is the identity.

Therefore $p = 1$ follows only from the conjunction:

1. positive multiplicative valuation;
2. a nonconstant profile;
3. scale-stationary path independence/idempotence;
4. exclusion of total erasure.

The third premise is stronger than ordinary coassociativity and is not a strict FIN theorem.

Without it, the continuous semigroup

$$p(t) = e^{-ct}, \qquad c \geq 0,$$

satisfies $p(t+s) = p(t)p(s)$ for every rate c. A rate modulus survives.

5 Round IV: ST2307–ST2321 — exact self-similarity

Let x be the normalized vector of twelve strict loop-product orbit values and define centered log coordinates

$$y_i = \log x_i - \frac{1}{12}\sum_j \log x_j.$$

Then T_p acts as $y \mapsto py$.

Theorem 5 (permuted self-similarity no-go)**.** *For a nonconstant positive* x*, if*

$$T_p(x) = Px$$

for a permutation matrix P *and* $p > 0$*, then* $p = 1$*.*

Proof. In centered log coordinates, $py = Py$. Since a permutation is orthogonal,

$$p\|y\| = \|Py\| = \|y\|.$$

Nonconstancy gives $\|y\| > 0$, hence $p = 1$. □

For $0 < p < 1$, normalized power refinement increases entropy and moves the finite profile toward uniformity. For $p > 1$, it lowers entropy and sharpens the largest orbit. Neither gives an exact nonconstant fixed shape.

The numerical idempotence residuals on the strict profile are

p	0	0.5	1	1.8	2
$\|T_p^2 x - T_p x\|_\infty$	0	2.43165	0	0.625304	0.467324.

One may preserve ordinary path composition with level-dependent exponents p_n, whose cumulative exponent is $\prod_n p_n$. This is exactly the escape already seen in FIN: one refinement rate or vertical mode per level. It is coherent but not predictive.

> Exact finite self-similarity selects identity or erasure. Nontrivial fractal evolution requires new level-dependent data.

6 Round V: ST2322–ST2336 — minimal ternary instrument

For

$$p_\varepsilon(x) = \frac{1}{8}(1 + \varepsilon x_1 x_2 x_3),$$

the minimal distinguishing observable is

$$Y = x_1 x_2 x_3.$$

Binary-cube character orthogonality proves that Y cannot be represented as a sum of one- and two-body functions. Hence interaction order three is minimal.

All pair marginals are independent of ε, so every pair-only instrument has Fisher information zero. In contrast,

$$\Pr(Y = y) = \frac{1 + \varepsilon y}{2}, \qquad y = \pm 1,$$

and

$$I(\varepsilon) = \frac{1}{1 - \varepsilon^2}.$$

The estimator

$$\widehat{\varepsilon} = \frac{1}{n}\sum_{r=1}^{n} Y_r$$

is unbiased with variance

$$\operatorname{Var}(\widehat{\varepsilon}) = \frac{1 - \varepsilon^2}{n},$$

which reaches the Cramér–Rao bound.

For $\varepsilon_0 = 0.75$ and declared error $\delta = 0.01$, the two-sided Hoeffding bound gives

$$n \geq \frac{2\log(2/\delta)}{\varepsilon_0^2},$$

so $n = 19$ events suffice.

The minimal exponential family carrying this statistic is

$$p_\theta(x) = \frac{\exp(\theta x_1 x_2 x_3)}{8\cosh\theta}, \qquad \varepsilon = \tanh\theta.$$

The likelihood flow

$$\dot{\theta} = \varepsilon_* - \tanh\theta$$

converges to $\theta = \operatorname{artanh}\varepsilon_*$ for a supplied target. At $\varepsilon_* = 0, \theta(0) = 0$, it remains zero. The model therefore makes the missing ingredient explicit rather than deriving it.

A Gibbs realization $H = -Jx_1x_2x_3$ has $\theta = \beta J$, but J, β, the sign, the preparation and the apparatus are new inputs.

7 Round VI: ST2337–ST2351 — synthesis

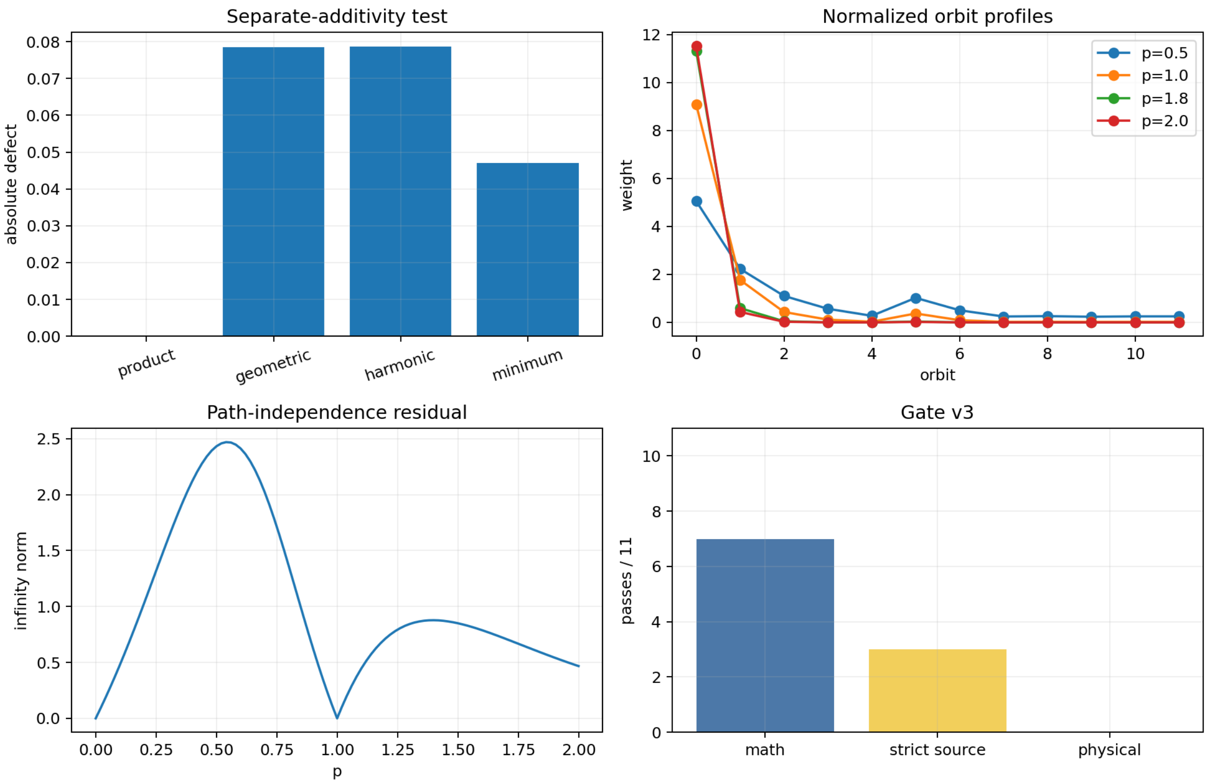

Figure 1: Separate-additivity counterrules; normalized orbit profiles; the identity-or-erasure residual; and gate-v3 separation of mathematical, strict-source and physical status.

Two conditional uniqueness routes now exist:

Route	Additional premises	Result
multilinear face lift	separate edge additivity + face universality	τ up to scale
fractal fixed point	multiplicativity + scale-stationary idempotence + non-erasure	$p = 1$, identity
ternary state	three-body source + parity instrument + polarity	non-Gaussian model
physical theory	one route + CA + local continuum + OA	conditional only

Neither uniqueness route is strict-derived. The fractal route is especially limited because its selected $p = 1$ map performs no nontrivial compression.

7.1 Gate v3

$\boxed{7/11 \text{ mathematical constructions}, \quad 3/11 \text{ strict-source rows}, \quad 0/11 \text{ physical rows.}}$

The strict rows are the loop-product field, its exact D_{12} structure, and the support-clique complex. Separate linearity, universality, idempotence, nontrivial refinement, irreducible ternary state, polarity, physical scale and OA remain unsourced.

8 Results ledger

Statement	Status	Scope
Dirichlet edge additivity forces face linearity	[Refuted]	category jump
weighted symmetry forces face universality	[Refuted]	$\mathrm{Aut}W = D_{12}$
continuous multiplicative valuations are powers	[Proven]	positive/continuous
idempotence selects $p = 0, 1$	[Proven]	nonconstant profile
information preservation selects $p = 1$	[Conditional]	added principle
$p = 1$ gives nontrivial compression	[Refuted]	identity map
nontrivial exact power self-similarity	[Impossible]	finite profile
pair instruments see hidden synergy	[Impossible]	zero Fisher info
parity is a sufficient ternary instrument	[Proven]	supplied model
strict FIN sources θ or J	[Not established]	state law absent
fundamental physical theory follows	[Refuted currently]	units/OA absent

9 Deepest surviving interpretation

> The strict FIN pair kernel possesses a rigorous loop-product flag lift, but neither edge composition nor its true symmetry selects a universal Hodge valuation. Imposing exact finite fractal self-similarity collapses the power law to identity or erasure, so it cannot generate a nontrivial hierarchy. Irreducible higher-order information is operationally accessible only after a three-body state statistic and joint instrument are added. FIN remains a finite Dirichlet–Markov–spectral information geometry; its current pair and refinement core cannot by itself become unique physical dynamics.

10 Next decisive direction

The loop-function and plain-coassociativity lanes should now be frozen. ST2352–ST2366 should test one genuinely strict non-Gaussian state law:

ST2352 inventory strict nonlinear state laws already present in the repository;

ST2353 test whether any contains an explicit three-body sufficient statistic;

ST2354 compute its connected third cumulant without fitted targets;

ST2355 test reflection covariance and nonzero polarity;

ST2356 distinguish state source from receiver/preparation;

ST2357 derive or reject its stationary measure;

ST2358 test Lyapunov stability and boundedness;

ST2359 test whether the pair kernel alone fixes its coefficient;

ST2360 construct $12 \to 24$ ternary transport;

ST2361 test $12 \to 24 \to 48$ path coherence;

ST2362 test compatibility with dual unitary/diffusive channels;

ST2363 test whether coarse records erase its signal;

ST2364 specify the parity-level operational record;

ST2365 audit dimensional and physical interpretation separately;

ST2366 issue a source theorem or a no-new-live-frontier certificate.

11 Reproducibility and nonclaims

Six result sets from `FIN_ST2262_ST2276_Results.json` through `FIN_ST2337_ST2351_Results.json` contain ninety packet hashes. `test_fin_st2262_st2351.py` verifies A3/A4 failures, the automorphism theorem, normalized-power identities, the self-similarity no-go, the minimal instrument and gate v3.

This report does not derive a strict three-body source, selector, polarity, nontrivial fractal scale law, unique Hodge dynamics, physical units, continuum gauge theory, apparatus, record, legacy completion, role transfer, QW-2191, Standard Model, gravity, L_{total}, or Theory of Everything.

References

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